ditronix.net April 21, 2026	Board Features	IPEM SIX	IPEM NB-IoT	IPEM Plus	EPEM 32	EPEM 36	IPEM PiHat	IPEM PiHat Lite	IPEM 32 (EOL)	IPEM 36 (EOL)
Released		**2026 Jun**	**2026 Jun**	**2026 May**	**2025 Dec**	**2025 Dec**	**2025 Apr**	**2025 Apr**	**2023 Feb**	**2023 Feb**
Processing	Espressif MCU Support	ESP32-C5	ESP32-C5	ESP32-C5	ESP32-C6	ESP32-C6	-	-	ESP32	ESP32
	Raspberry Pi Support (40 Pin)	-	-	-	-	-	Yes	Yes	-	-
AC Monitoring	ATM90E32 Energy Monitor Device	Yes	-	-	Yes	-	-	-	Yes	-
	ATM90E36 Energy Monitor Device	-	Yes	Yes	-	Yes	Yes	Yes	-	Yes
	ATM Fast DMA or SPI Modes	-	Yes	Yes	-	Yes	Yes	Yes	-	Yes
	ATM90E36 DMA Raw Data Dumping	-	Yes	Yes	-	Yes	Yes	Yes	-	Yes
	ATM90E36 DMA Data Acquisition	-	Yes	Yes	-	Yes	Yes	Yes	-	Yes
	Total Harmonic Distortion (THD)	-	Yes	Yes	-	Yes	Yes	Yes	-	Yes
	Fast Fourier Transform (FFT)	-	Yes	Yes	-	Yes	Yes	Yes	-	Yes
	Discrete Fourier Transform (DFT)	-	Yes	Yes	-	Yes	Yes	Yes	-	Yes
	Dynamic Range	6000:1	6000:1	6000:1	6000:1	6000:1	6000:1	6000:1	6000:1	6000:1
	Active Energy Accuracy	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%
	Reactive Energy Accuracy	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%
Configuration	Board Configuration	Jumper	Jumper	Jumper	Jumper	Jumper	DIL Switch	DIL Switch	Solder Pad	Solder Pad
	Stackable Boards (Extra CT Inputs)	-	-	-	-	-	-	Yes	-	-
Connections	Connections 3.5mm Jack (CT)	-	-	-	Yes	Yes	Yes	Yes	Yes	Yes
	Terminal Blocks	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Current Inputs	CT Clamp Inputs (Typically 0 to 100A)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	YHDC SCT013-000 100A / 50mA 1:2000	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Non YDHC Current Clamp Inputs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	-	-
	Rogowski Coil Inputs (Typically > 100A)	-	Yes	Yes	Yes	Yes	Yes	Yes	-	-
	Rogowski Current Transfomer PA3202NL	-	Yes	Yes	Yes	Yes	Yes	Yes	-	-
	AC Voltage Inputs	1, 2 or 3	1, 2 or 3	1, 2 or 3	1, 2 or 3	1, 2 or 3	1, 2 or 3	1, 2 or 3	1, 2 or 3	1, 2 or 3
	AC Current Inputs	6 (2 x 3)	4 (3 + N)	4 (3 + N)	3	4 (3 + N)	4 (3 + N)	4 (3 + N)	3	4 (3 + N)
	Neutral Current Monitoring	-	Yes	Yes	-	Yes	Yes	Yes	-	Yes
AC Input	AC Sense Inputs (Low Voltage)	3	3	3	3	3	3	3	3	3
	AC Inputs Configurable (1, 2 or 3)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Board Power	AC Power Input SMPS (5 to 20V RMS)	Yes	Yes	Yes	-	-	-	-	Yes	Yes
	DC Power Input SMPS (5 to 20V)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	DC USB Power Option	Yes	Yes	Yes	Yes	Yes	-	Yes	Yes	Yes
AC Phases	Single Phase (x3) Monitoring	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Dual or Split Phase Monitoring	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Three Phase three-wire (3P3W, Y or Δ)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Three Phase four-wire (3P4W, Y0)	-	Yes	Yes	-	Yes	Yes	Yes	-	Yes
IoT	Home Assistant Support	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	ESPHome Support	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Tasmota Support	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Domoticz Support	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Cloud Based Services	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	MQTT Support	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Connectivity	LoRa Support	-	-	Yes	-	-	-	-	-	-
	NB-IoT GPRS SIM7000G	-	Yes	-	-	-	-	-		
	Radio Modules - 2.4 GHz WiFi	Yes	Yes	Yes	Yes	Yes	Raspberry Pi	Raspberry Pi	Yes	Yes
	Radio Modules - 5 GHz WiFi	Yes	Yes	Yes	-	-	Raspberry Pi	Raspberry Pi	-	-
	Radio Modules - Bluetooth 2.4 GHz	Yes	Yes	Yes	Yes	Yes	Raspberry Pi	Raspberry Pi	Yes	Yes
	Radio Modules - Zigbee 2.4 GHz	Yes	Yes	Yes	Yes	Yes	-	-	-	-
	Radio Modules - Thread 2.4 GHz	Yes	Yes	Yes	Yes	Yes	-	-	-	-
	Radio Modules - LoRa 850 to 930 MHz	-	-	Yes	-	-	-	-	-	-
	Radio Modules - LoRa SX1262 (FCC/CE)	-	-	Yes	-	-	-	-	-	-
	RS-485	Yes	Yes	Yes	Yes	Yes	-	-	-	-
	Ethernet (W5500)	-	-	-	Yes	Yes	Raspberry Pi	Raspberry Pi	-	-
Interfaces	Galvanically Isolated Opto Output (50mA)	-	Yes	Yes	-	-	-	-	-	-
	Realtime Clock Battery Option (CR2032)	-	-	Yes	Yes	Yes	Yes	-	-	-
	OLED Display Interface I2C	Yes	-	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	RS485 / DMX512 Interface	Yes	Yes	Yes	Yes	Yes	-	-	-	-
	Arduino & PlatformIO Devkit Board	ESP32-C5	ESP32-C5	ESP32-C5	ESP32-C6	ESP32-C6	-	-	ESP32	ESP32
	USB Interface Micro (CH340C UART)	-	-	-	-	-	-	-	Yes	Yes
	USB Interface Type C (CH343P UART)	Yes	Yes	Yes	Yes	Yes	-	Power Only	-	-
	Expansion Interface Panel I2C	Yes	Yes	Yes	-	-	-	-	-	-
	Expansion Interface SPI	Yes	Yes	Yes	-	-	-	-	-	-
	OneWire Support Interface (DS18B20)	Yes	Partly	Partly	-	-	-	-	-	-
	DAC Output	-	MCU	MCU	-	-	MCP4728	-	-	-
	FET Output	?	Yes	Yes	Yes	Yes	-	-	-	-
	PWM/GPO Out	?	Yes	Yes	Yes	Yes	-	-	-	-
Other	MicroSD Card Capabilty	Yes	Yes	Yes	-	-	Raspberry Pi	Raspberry Pi	-	-
	Internal EEPROM (24C64 64k)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Realtime Clock DS3231SN	-	-	Yes	Yes	Yes	Yes	-	-	-
	Realtime Clock Battery Backup	-	-	Yes	Yes	Yes	Yes	-	-	-
	ADC Inputs ADS1115 16bit	Yes	Yes	Yes	-	-	Yes	-	-	-
	Analogue Temperature Sensor NTC	-	-	-	-	-	-	-	Yes	Yes
	Digital Temperature Sensor TMP102	Yes	Yes	Yes	Yes	Yes	Yes	-	-	-
Mechanical	DIN Encloure Mount Option	Yes	Yes	Yes	-	-	-	-	-	-
	Display Panel Option	Yes	Yes	Yes	-	-	-	-	-	-
Applications	Home Monitoring	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Offgrid Monitoring	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Solar (Farm) Monitoring	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Energy Gateways	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Data Concentrator	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Electric Metering	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	NLM Data Monitoring	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
E&OE	Industrial and Commercial Monitoring	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Farming and Agriculture Monitoring	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes